Week 6–8 References

Domain 3: Data Transformation & Modeling (22%)

Databricks Certified Data Engineer Associate

Verified: August 2026

Topics: Medallion · PySpark · Spark SQL · JOINs
Window Functions · Arrays · Tuning · Gold Layer Objects

Quick Navigation

1. Medallion Architecture (Bronze-Silver-Gold)
2. PySpark DataFrame API
3. Spark SQL
4. JOIN Operations & Optimization
5. Aggregations & Deduplication
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7. Array & Nested Operations
8. Spark Performance Tuning
9. Gold Layer Objects (Views, MVs, ST)
10. Data Quality & Validation
11. Best Practices & Patterns
12. Release Notes & Updates

Medallion Architecture

Layer	URL
Medallion Overview	https://docs.databricks.com/en/lakehouse/index.html
Bronze Layer (Raw)	https://docs.databricks.com/en/lakehouse/index.html
Silver Layer (Cleaned)	https://docs.databricks.com/en/lakehouse/index.html
Gold Layer (Analytics)	https://docs.databricks.com/en/lakehouse/index.html
Bronze → Silver → Gold	https://docs.databricks.com/en/lakehouse/index.html
Change Data Feed in Bronze	https://docs.databricks.com/en/delta/change-data-feed.html
Incremental Processing in Silver	https://docs.databricks.com/en/lakehouse/index.html
Data Quality in Silver	https://docs.databricks.com/en/lakeflow/pipelines/expectations.html

PySpark DataFrame API

Topic	URL
DataFrame Basics	https://docs.databricks.com/en/pyspark/dataframes.html
select() & selectExpr()	https://docs.databricks.com/en/pyspark/dataframes.html
withColumn() & withColumnRenamed()	https://docs.databricks.com/en/pyspark/dataframes.html
filter() & where()	https://docs.databricks.com/en/pyspark/dataframes.html
groupBy() & agg()	https://docs.databricks.com/en/pyspark/dataframes.html
drop() & dropDuplicates()	https://docs.databricks.com/en/pyspark/dataframes.html
orderBy() & limit()	https://docs.databricks.com/en/pyspark/dataframes.html
cache() & persist()	https://docs.databricks.com/en/pyspark/dataframes.html
write.mode() & write.format()	https://docs.databricks.com/en/pyspark/dataframes.html

Spark SQL

Topic	URL
SELECT & WHERE	https://docs.databricks.com/en/sql/language-manual/sql-ref-syntax-dml-select.html
GROUP BY & HAVING	https://docs.databricks.com/en/sql/language-manual/sql-ref-syntax-dml-select.html
ORDER BY & LIMIT	https://docs.databricks.com/en/sql/language-manual/sql-ref-syntax-dml-select.html
WITH (CTE)	https://docs.databricks.com/en/sql/language-manual/sql-ref-syntax-dml-select.html
Subqueries & Derived Tables	https://docs.databricks.com/en/sql/language-manual/sql-ref-syntax-dml-select.html
EXISTS / NOT EXISTS	https://docs.databricks.com/en/sql/language-manual/sql-ref-syntax-dml-select.html
CASE WHEN THEN	https://docs.databricks.com/en/sql/language-manual/sql-ref-syntax-dml-select.html

JOIN Operations & Optimization

Topic	URL
INNER JOIN	https://docs.databricks.com/en/sql/language-manual/sql-ref-syntax-dml-select-join.html
LEFT OUTER JOIN	https://docs.databricks.com/en/sql/language-manual/sql-ref-syntax-dml-select-join.html
FULL OUTER JOIN	https://docs.databricks.com/en/sql/language-manual/sql-ref-syntax-dml-select-join.html
CROSS JOIN	https://docs.databricks.com/en/sql/language-manual/sql-ref-syntax-dml-select-join.html
BROADCAST JOIN	https://docs.databricks.com/en/sql/language-manual/sql-ref-syntax-dml-select-join.html
Join Hints /*+ BROADCAST() */	https://docs.databricks.com/en/sql/language-manual/sql-ref-syntax-dml-select-join.html
spark.sql.autoBroadcastJoinThreshold	https://docs.databricks.com/en/sql/conf.html
Shuffle Join	https://docs.databricks.com/en/sql/conf.html
UNION / UNION ALL	https://docs.databricks.com/en/sql/language-manual/sql-ref-syntax-dml-select-set-ops.html

Aggregations & Deduplication

Topic	URL
COUNT, SUM, AVG, MIN, MAX	https://docs.databricks.com/en/sql/language-manual/sql-ref-functions-builtin.html
APPROX_COUNT_DISTINCT	https://docs.databricks.com/en/sql/language-manual/sql-ref-functions-builtin.html
STDDEV, VARIANCE	https://docs.databricks.com/en/sql/language-manual/sql-ref-functions-builtin.html
COLLECT_LIST, COLLECT_SET	https://docs.databricks.com/en/sql/language-manual/sql-ref-functions-builtin.html
GROUP BY Clause	https://docs.databricks.com/en/sql/language-manual/sql-ref-syntax-dml-select.html
HAVING Clause	https://docs.databricks.com/en/sql/language-manual/sql-ref-syntax-dml-select.html
DISTINCT	https://docs.databricks.com/en/sql/language-manual/sql-ref-syntax-dml-select.html
dropDuplicates() in PySpark	https://docs.databricks.com/en/pyspark/dataframes.html

Window Functions

Topic	URL
OVER Clause	https://docs.databricks.com/en/sql/language-manual/sql-ref-syntax-dml-select.html
PARTITION BY & ORDER BY	https://docs.databricks.com/en/sql/language-manual/sql-ref-syntax-dml-select.html

ROW_NUMBER()	https://docs.databricks.com/en/sql/language-manual/sql-ref-functions-builtin.html
RANK() & DENSE_RANK()	https://docs.databricks.com/en/sql/language-manual/sql-ref-functions-builtin.html
LAG() & LEAD()	https://docs.databricks.com/en/sql/language-manual/sql-ref-functions-builtin.html
FIRST_VALUE() & LAST_VALUE()	https://docs.databricks.com/en/sql/language-manual/sql-ref-functions-builtin.html
Running Sum: SUM() OVER	https://docs.databricks.com/en/sql/language-manual/sql-ref-functions-builtin.html
Top N per Group Pattern	https://docs.databricks.com/en/sql/language-manual/sql-ref-functions-builtin.html

Array & Nested Operations

Topic	URL
EXPLODE()	https://docs.databricks.com/en/sql/language-manual/sql-ref-functions-builtin.html
FLATTEN()	https://docs.databricks.com/en/sql/language-manual/sql-ref-functions-builtin.html
array_contains()	https://docs.databricks.com/en/sql/language-manual/sql-ref-functions-builtin.html
collect_list() & collect_set()	https://docs.databricks.com/en/sql/language-manual/sql-ref-functions-builtin.html
Struct Data Type	https://docs.databricks.com/en/sql/language-manual/sql-ref-data-types-struct.html
named_struct()	https://docs.databricks.com/en/sql/language-manual/sql-ref-functions-builtin.html
get_json_object()	https://docs.databricks.com/en/sql/language-manual/sql-ref-functions-builtin.html
from_json() & to_json()	https://docs.databricks.com/en/sql/language-manual/sql-ref-functions-builtin.html

Spark Performance Tuning

Topic	URL
spark.sql.shuffle.partitions	https://docs.databricks.com/en/sql/conf.html
spark.default.parallelism	https://docs.databricks.com/en/sql/conf.html
spark.sql.autoBroadcastJoinThreshold	https://docs.databricks.com/en/sql/conf.html
Executor & Driver Memory	https://docs.databricks.com/en/compute/configure.html
Partitioning Strategy	https://docs.databricks.com/en/optimization/index.html
Caching Strategy	https://docs.databricks.com/en/pyspark/dataframes.html
Predicate Pushdown	https://docs.databricks.com/en/optimization/index.html
Spark UI - Jobs/Stages/Tasks	https://docs.databricks.com/en/clusters/monitor/spark-ui.html
Data Skew Detection	https://docs.databricks.com/en/clusters/monitor/spark-ui.html
Adaptive Query Execution (AQE)	https://docs.databricks.com/en/optimizations/adaptive-query-execution.html

Gold Layer Objects

Topic	URL
CREATE VIEW	https://docs.databricks.com/en/sql/language-manual/sql-ref-syntax-ddl-create-view.html
CREATE MATERIALIZED VIEW	https://docs.databricks.com/en/sql/language-manual/sql-ref-syntax-ddl-create-materialized-view.html
REFRESH MATERIALIZED VIEW	https://docs.databricks.com/en/sql/language-manual/sql-ref-syntax-ddl-alter-materialized-view.html
CREATE STREAMING TABLE	https://docs.databricks.com/en/lakeflow/pipelines/streaming-tables.html
View vs Materialized View	https://docs.databricks.com/en/sql/language-manual/sql-ref-syntax-ddl-create-view.html

Data Quality & Validation

Topic	URL
Expectations in Lakeflow	https://docs.databricks.com/en/lakeflow/pipelines/expectations.html
CONSTRAINT	https://docs.databricks.com/en/lakeflow/pipelines/expectations.html
NOT NULL, Range Validation	https://docs.databricks.com/en/lakeflow/pipelines/expectations.html
Quarantine Pattern	https://docs.databricks.com/en/lakeflow/pipelines/expectations.html
Dead Letter Queue	https://docs.databricks.com/en/lakeflow/pipelines/expectations.html

Best Practices & Patterns

Pattern	URL
Minimize Shuffles	https://docs.databricks.com/en/optimization/index.html
Push Filters Early	https://docs.databricks.com/en/optimization/index.html
Broadcast Small Tables	https://docs.databricks.com/en/sql/language-manual/sql-ref-syntax-dml-select-join.html
Incremental Merge	https://docs.databricks.com/en/delta/index.html
SCD Type 2	https://docs.databricks.com/en/lakehouse/index.html
Late Arriving Facts	https://docs.databricks.com/en/lakehouse/index.html

Summary: 95+ official Databricks documentation URLs covering Domain 3 (22% of exam). Verified August 2026 against May 2026 Exam Guide.